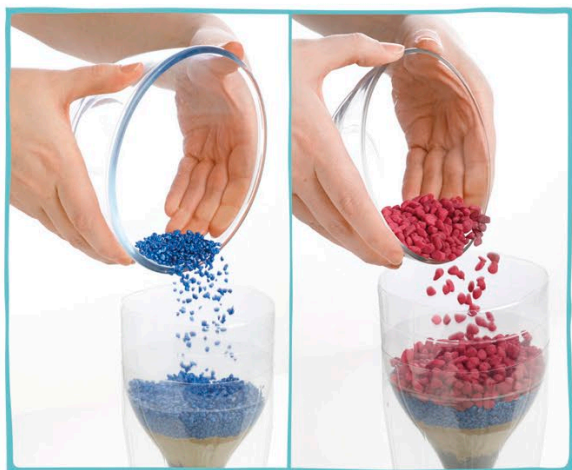




3 Put the top part of the bottle upside down into the bottom part of the bottle. Add a layer of charcoal, about $\frac{1}{2}$ in (1cm) deep, on top of the cotton balls. If your charcoal is in big chunks, crush it first to break it into smaller pieces.



4 Add sand about $\frac{3}{4}$ in (2cm) deep. Press it hard with your fingertips to push down the sand and the charcoal beneath it. These packed layers will slow down water flow and trap lots of dirt.



5 Next, add a $\frac{1}{2}$ in (1cm) layer of the small gravel. On top of that, add a $\frac{3}{4}$ in (2cm) layer of medium-sized gravel. You must pack these two layers as firmly as you can, just as you did with the sand and the charcoal.

The more layers you add, the cleaner your water will get as it filters down.

6 Finally, add the small pebbles. Make sure you completely cover the top of the gravel. See how the gaps between the particles of each layer have grown bigger and bigger toward the top. Now that your water filter is ready to put to work, you need to make some dirty water!

